Area Between Curves

Section 1: Area Between Curves Integrated with Respect to x

1 What is the area of the region enclosed by $y = x^2$ and $y = 4$?

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 16

d. 8

2 Find the area between $y = x$ and $y = x^2$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{1}{12}$

3 Which integral represents the area enclosed by $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{4}$?

a. $\int_0^{\pi/4} (\cos x - \sin x)dx$

b. $\int_0^{\pi/4} (\sin x - \cos x)dx$

c. $\int_0^{\pi/4} (\cos x + \sin x)dx$

d. $\int_0^{\pi/2} \cos x dx$

Section 2: Area Between Curves Integrated with Respect to y

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4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$.

a. $\frac{9}{2}$

b. $\frac{16}{3}$

c. $\frac{7}{2}$

d. 6

5 When is it advantageous to integrate with respect to y rather than x?

a. When the boundary curves are expressed as $x = f(y)$ and solving for y requires multiple piecewise branches

b. When the area is negative

c. Always for quadratic curves

d. Never on the AP Exam

6 Find the area bounded by $x = 0, y = 1, y = 3,$ and $x = \frac{1}{y}$.

a. $\ln 3$

b. $\ln 2$

c. 3

d. 2

Section 3: Intersecting Curves with Multiple Regions

7 The graphs of $y = x^3$ and $y = x$ enclose how many symmetric regions?

a. 2 symmetric regions

b. 1 region

c. 3 regions

d. Infinitely many

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8 What is the total area enclosed by $y = x^3$ and $y = x$?

a. $\frac{1}{2}$

b. $\frac{1}{4}$

c. 0

d. 1

9 Why does $\int_{-1}^{1}(x - x^3)dx = 0$ NOT represent the enclosed area?

a. The integrand is odd, causing positive and negative signed areas to cancel out

b. The curves do not intersect at 0

c. The integral diverges

d. The bounds are incorrect

Section 4: Area Partitions & Horizontal/Vertical Lines

10 The line $x = k$ divides the region between $y = x^2$ and the x-axis from $x = 0$ to $x = 2$ into two equal areas. Find k.

a. $\sqrt[3]{4}$

b. 1

c. $\sqrt{2}$

d. $\frac{4}{3}$

11 Find the area enclosed by $y = e^x$, $y = e^{-x}$, and the vertical line $x = 1$.

a. $e + \frac{1}{e} - 2$

b. $e - 1$

c. $e + \frac{1}{e}$

d. $2e - 2$

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12 Find the area of the region bounded by $y = \sqrt{x}$, the line $y = 2$, and the y-axis.

a. $\frac{8}{3}$

b. 4

c. $\frac{4}{3}$

d. 2

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